Vehicle Rental System

Introduction

- A Vehicle Rental Management System is a software application designed to streamline the process of renting vehicles, like cars and bikes, to customers for a specified period.
- Rental businesses need an efficient system to track vehicle availability, manage customers, and record rental transactions.
- Traditionally, vehicle rental operations are managed manually using paper records or spreadsheets, which can lead to inefficiency and errors.
- Manual systems make it challenging to quickly check vehicle availability or calculate rental charges accurately.
- A software-based rental management system automates operations, reducing human effort.
- Automation enhances data organization, speeds up service, improves the accuracy of records, and boosts customer satisfaction.

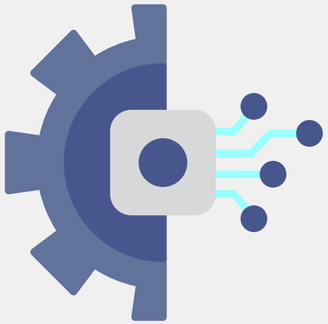
Problem Statement

1: Challenges of Manual Rental Work

- Difficulty in identifying available or rented vehicles
- Risk of records being lost, copied incorrectly, or filled in inaccurately
- Manual billing can result in errors
- Lack of a centralized location for vehicle and rental details
- Time-consuming process for finding or updating information

2: Issues with Tracking and Billing

- Slow and unclear tracking of vehicle availability
- Potential for paper records to be mixed up or written incorrectly
- Extra time required for billing, leading to possible errors
- No shared database for vehicle and rental data
- Updates are tiresome and slow without automation



3: Benefits of an Automated System

- Simplifies and clarifies rental tracking
- Reduces the risk of losing or duplicating records
- Minimizes billing mistakes with automated calculations
- Provides a shared space for all vehicle and rental information
- Speeds up updates, making the rental process faster and more reliab

Objectives

User-friendly System

The system aims to provide an intuitive interface for seamless vehicle management.

Vehicle Tracking

Enables real-time tracking of vehicle availability and rental status for users.

OOP Implementation

Focus on utilizing Object Oriented Programming principles for efficient code structure.

Customer Records

Keeps rental history and customer details organised.

Technologies and Tools Used

C++ Utilizing Modern Standards for Core Project Logic

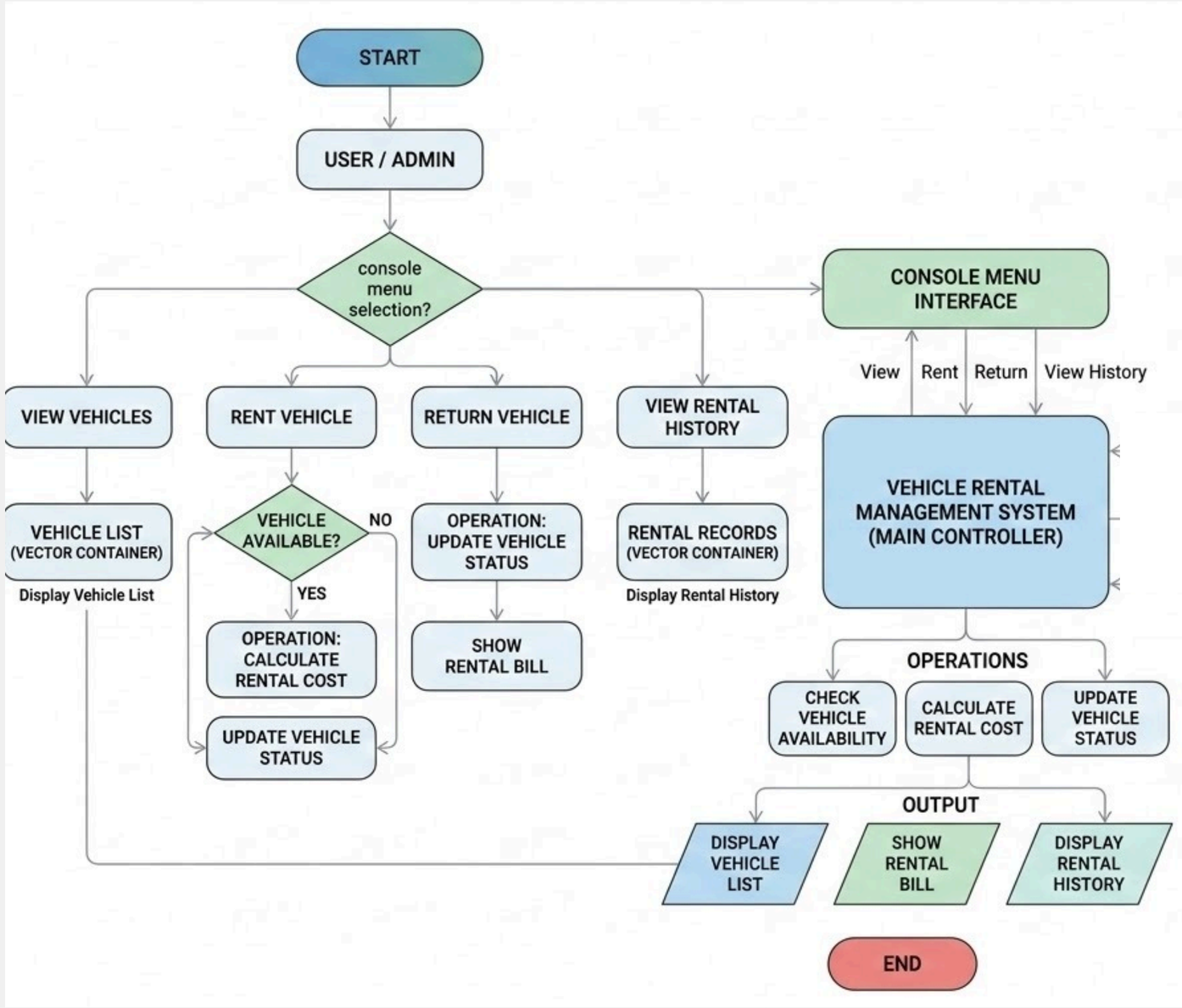
- Employ a modern C++ approach for robust project logic,
- Emphasizing an object-based structure and seamless feature management.

Data Handling with STL Tools

- Utilize STL tools such as dynamic containers, file streams, and straightforward CSV storage to effectively manage data.

System Architecture

Visual representation of the vehicle class



This UML diagram illustrates the structural design of the Vehicle Rental Management System. It highlights the Vehicle class as the base class, encapsulating common attributes and methods shared by derived classes.

OOP Concepts in Vehicle Rental System

Encapsulation

Essential to hide data and provide controlled access via methods and functions.
Vehicle attributes like isAvailable are kept protected.

Inheritance

The Car and Bike classes "inherit" the ID, Brand, and Rate logic from the Vehicle class.

Polymorphism

The system decides at runtime whether to display Car-specific details (Seats/Transmission) or Bike-specific details (Complimentary Helmet).

Abstraction

Abstraction focuses on hiding the complex background details and showing only the necessary interface to the user.
The Vehicle class is an Abstract Base Class because of the "Pure Virtual Function"

Key Features of the Vehicle Rental System

Add Vehicles

Easily incorporate new vehicles into the system.

Display Info

Showcase detailed vehicle information promptly.

Rent Vehicles

Streamlined process for renting vehicles quickly.

Cost Calculation

Automated calculations for accurate billing provided.

Process Returns

Efficient management of vehicle returns ensured.

Rental History

Comprehensive records of all past rentals managed.

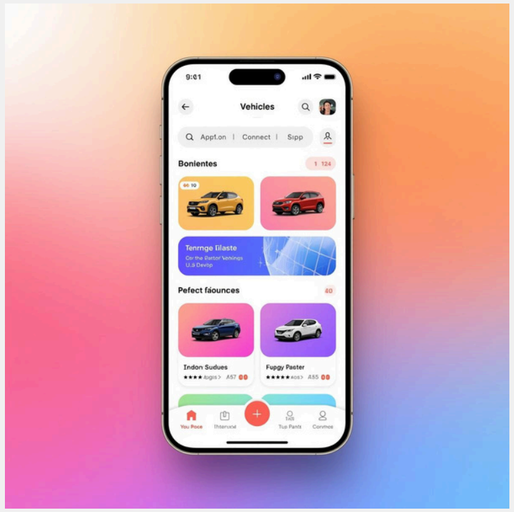
Results and Program Output Overview

```
==========================================
      SMART VEHICLE RENTAL SYSTEM (OOPS)
==========================================
1. Admin Login
2. Customer Portal
Choice: 2

>> CUSTOMER PORTAL
1. View Available Vehicles
2. Rent a Vehicle
3. Return a Vehicle
4. Exit System
Select Action: 1

CURRENT VEHICLE FLEET

[1001] Toyota Fortuner    | Rate: ₹4500/day | AVAILABLE
      Specs: 7 Seater | Auto Transmission

[1002] Mahindra Thar      | Rate: ₹3500/day | AVAILABLE
      Specs: 4 Seater | Manual Transmission

[1004] Hyundai Creta      | Rate: ₹3000/day | AVAILABLE
      Specs: 5 Seater | Auto Transmission

[1005] Maruti Swift       | Rate: ₹1500/day | AVAILABLE
      Specs: 5 Seater | Manual Transmission

[1007] KTM Duke 390       | Rate: ₹1500/day | AVAILABLE
      Specs: Sports Engine | Complimentary Helmet

[1008] Yamaha R15         | Rate: ₹1100/day | AVAILABLE
      Specs: Sports Engine | Complimentary Helmet

[1009] TVS Apache         | Rate: ₹900/day | AVAILABLE
      Specs: Sports Engine | Complimentary Helmet

[1010] Suzuki Hayabusa    | Rate: ₹7500/day | AVAILABLE
      Specs: Sports Engine | Complimentary Helmet

[1011] Skoda Kylaq        | Rate: ₹2500/day | AVAILABLE
      Specs: 5 Seater | Manual Transmission

[1012] BMW                | Rate: ₹3000/day | AVAILABLE
      Specs: 4 Seater | Auto Transmission

[1013] RE                 | Rate: ₹1200/day | AVAILABLE
      Specs: Sports Engine | Complimentary Helmet
```

Displaying Key Rental Information

The system generates a comprehensive vehicle list, presenting detailed information including availability status and rental costs, facilitating seamless management and user interaction within the rental process.

Future Enhancements for Vehicle Rental System



User-Friendly GUI

Enhancing user experience with intuitive design.

Database Integration

Ensuring reliable data storage and access.

Thank You

This project successfully implemented a vehicle rental management system using C++. Key concepts of Object Oriented Programming were applied, resulting in efficient management of rentals and improved user experience.